

Re-spend the Tokens Where the Model Already Looks: Depth-Profile Re-ranking for Small-Object VQA

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Abstract

Vision-language models (VLMs) fail on small objects because the evidence is never encoded, not because attention is misdirected. On V*Bench, 87% and 84% of two models’ errors are fixed by spending the same visual-token budget on the evidence region instead of the whole image; an oracle crop is flat at 96–100% across a threshold of ≈ 0.25 merged tokens of target extent below which uniform encoding falls under the 25% chance level. Seven internal interventions cannot cross that threshold: amplifying attention on exactly the right cells recovers 16% of what re-encoding those cells recovers, because the oracle set is one cell of 294. We give a method that does cross it. Depth-Profile Re-ranking reads the language model’s attention over image cells at all 28 layers, ranks cells with a small learned head, and crops once. Against the equal-compute baseline of encoding a larger image, it is the only arm in our study that wins, and it wins on two architectures and two benchmarks on single-object questions (+15.7, +11.3, +7.5, +8.5 pp, all CIs clear); the standard argmax read-out clears in no cell. Along the way we show the deployed read-out is wrong on four separable axes — it averages layers where a max is worth +7 pp, reads the attention weight rather than $\alpha\|v\|$, averages all heads, and fails entirely unless the prompt ends at the answer-emission point — and that ranking visual tokens by layer-2 attention is worse than ranking them at random on two models and three benchmarks. We report the method’s three measured limits (relational questions, existence questions, one architecture whose layers agree), a ≈ 91 -parameter readable head at parity with the learned one, a label-free rule within 3 pp of it, and every negative — including four of our own claims rejected by replication and a six-week-old negative overturned by one line of a prompt.

1 Introduction

A VLM that is asked “what colour is the glove?” and answers wrongly can fail in two ways: it can attend to the wrong place, or it can attend to the right place and find nothing there. A growing family of methods assumes the first — highlighting evidence pixels (Liu et al., 2026), rescaling attention heads (Lu et al., 2026), supervising attention during training (Esmaeilkhani & Latecki, 2026) — and all report gains. None measures how large the queried object is. We do, and on the benchmark built for small objects the second failure dominates: the model localises, the region is causally load-bearing, and the tokens covering it carry nothing the answer needs.

The practical consequence is that the only fix is to re-spend the token budget: encode the region the model already points at, at the resolution the budget affords. That is not a new idea — attention-guided cropping is a family of its own (Zhang et al., 2025; Shen et al., 2024; Anonymous, 2026a) — but as far as we can find, no member of it has been scored against the one baseline that decides whether it earns its second forward pass: spend the same tokens on a larger image. When we run that comparison, every published policy we test breaks even or loses (§G). Ours is the only arm that wins, and the reason is a read-out, not a policy.

Contributions.

- A method that beats equal compute (§3). Depth-Profile Re-ranking (DPR) ranks image cells by their full 28-layer attention profile and crops once. On single-object questions it beats a uniform image at the same token count on V*Bench and HR-Bench 4k, on Qwen3-VL-2B and Qwen2-VL-7B: +15.7, +11.3, +7.5, +8.5 pp, all confidence intervals clear of zero. The deployed argmax clears in no cell. Zero-shot to HR-Bench 4k with nothing refitted.
- The strong baseline it is measured against (§4). The standard read-out — mean over a late block of layers, mean over heads, argmax — is wrong on four axes, each fixable in one line. We port the fixes from LLM work (max over layers (McDanel et al., 2026), $\alpha\|v\|$ (Kobayashi et al., 2020), head selection (Lu et al., 2026)) and find they do not stack; the layer set does. A label-free rule — max over the layers where attention is question-conditioned — is within 3.7/2.7 pp of the learned head. Ranking tokens by layer-2 attention is **below random** on two models and three benchmarks.
- Why it works (§5). Below ≈ 0.25 merged tokens the evidence is not encoded: 87%/84% of errors are oracle-fixable, the oracle is flat across the threshold, and seven internal interventions fail. Cropping restores an answer-formation step at layer 21, only where the crop delivers.
- Where it stops (§6). Three boundaries, each with a mechanism and each found by our own controls: relational questions (the evidence set is a union $7.9\times$ larger), existence questions (a crop is evidence of absence; POPE’s oracle is circular), and one architecture whose layers agree with each other. The learned read-out is task-specific: 3,000 public boxed items from a text-reading task train a head worse than no head.

Everything is pre-registered where a selection was possible, replicated on a second model before it is claimed, and reported with the negatives at equal weight (§9, App. M).

2 Setup

Benchmarks. V*Bench (Wu & Xie, 2024): 191 four-way questions on ~ 1500 – 2000 px images whose targets are small by design; ground-truth boxes; 115 single-object (direct_attributes) and 76 relational (relative_position). HR-Bench 4k (Wang et al., 2024): 800 rows / 200 instances at 4032px, no boxes, evaluated per row and under CircularEval (all four option cycles correct). POPE (Li et al., 2023) with RePOPE labels (Neuhaus & Hein, 2025) and COCO boxes (482 relabelled, 812 dropped as ambiguous) for the scope test of §6; MMBench (Liu et al., 2024) for pruning only; 3,000 TextVQA items (Singh et al., 2019) with evidence boxes from Visual-CoT (Shao et al., 2024) for §6. Public benchmarks only; nothing constructed.

Models. Qwen3-VL-2B and Qwen2-VL-7B (all end-task claims); LLaVA-NeXT-7B and LLaVA-OneVision-7B for read-out and mechanism claims (their image_newline separators are stripped and the grid rebuilt, App. A). Frozen weights throughout; bf16.

Budget accounting. Every arm’s visual-token count is measured from the processor’s image_grid_thw, never computed; a contrast whose arms drift by more than 10% is void. The method costs two 300-token passes; its baseline is one 600-token pass (“uniform@600”, the bar). Reported drifts are 0.1–1.3%.

Read-out and statistics. Answers are read from the option logits, no generation. Confidence intervals are 95% item-level bootstraps (≥ 4000 resamples); a claim is made only if its CI clears zero on both Qwen models. Every learned quantity is out-of-fold, GroupKFold(5) grouped by item so an item’s ~ 300 cells never straddle folds; selections are made on training folds only. The run-to-run floor on top-1 coverage is 1–2.5 pp (App. A).

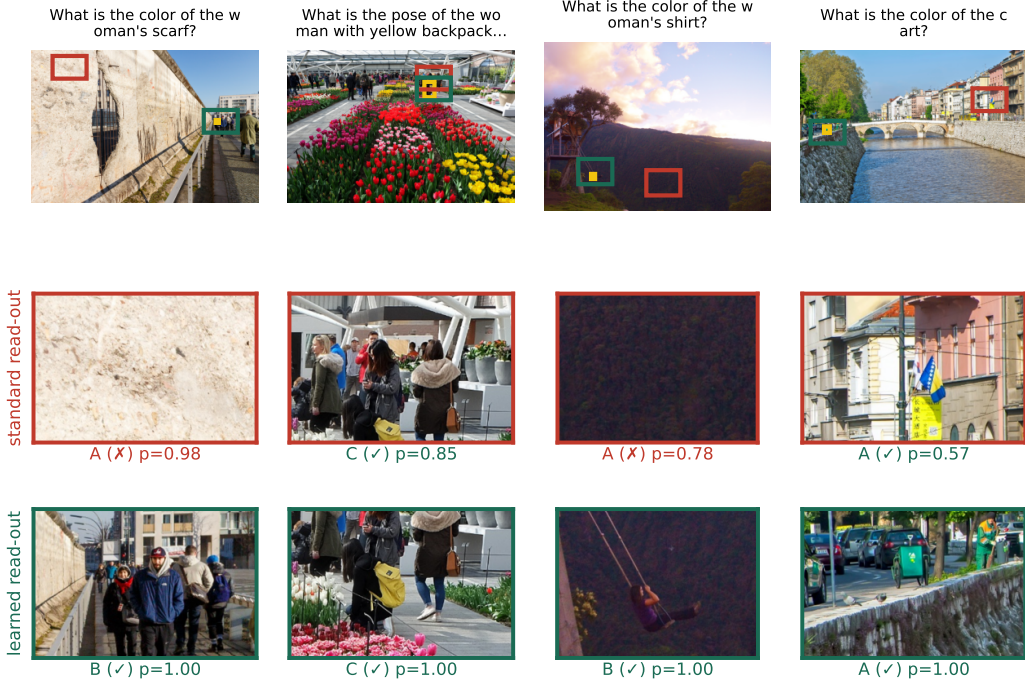


Figure 1: The method, on real items. Top: the source image with the ground-truth evidence box. Below: the 300-token crops the model is actually fed under the deployed argmax ("standard") and under DPR ("learned"), with the model's actual answer under each. The learned head's crop contains the evidence; the argmax crop lands on a distractor with high attention. Every crop and answer here comes from a forward pass, not an illustration.

3 The method: Depth-Profile Re-ranking

Localise. Encode the image at $B_0=300$ merged tokens with the full question. From that pass, take the final token's attention over the n image cells at every one of the 28 language-model layers, head-averaged and sum-normalised per layer: a $28 \times n$ profile. The prompt must end at the answer-emission point (§4); this is the whole of what makes the map usable.

Rank. Score each cell with 65 features from that pass alone: its 28 depth values, its 28 within-layer ranks, a 3×3 neighbourhood mean, the deployed block-mean, and seven geometry terms including explicit last-column / last-row / first-column indicators so the head can learn the serialisation sink (App. D) if that is what matters. The scorer is a gradient-boosted regressor (depth 4, 150 trees) trained to predict coverage — the fraction of the evidence box a window centred on the cell would contain — out-of-fold, grouped by item. Border cells are ring-masked, as in the deployed proposer.

Spend. Crop a window of side W of the image around the top cell, re-encode at 300 tokens, answer. $W=0.25$ on V*Bench (held-out validated: folds pick it 99/100, in-sample 71.7% $\rightarrow$ held-out 71.6%); $W=0.15$ unswept on HR-Bench 4k.

The gap it exploits. On V*Bench the deployed argmax covers the evidence 40.5% of the time; oracle selection among the same map's top-5 covers 61.5%; the ground-truth cell already sits in the top 3.4% of cells. The ranking has the answer and the max throws it away, usually to a sink or a distractor (Fig. 1).

Table 1: DPR against equal compute on single-object questions. Bar = one uniform image at the method’s total token count. Argmax = the deployed block-mean read-out feeding the same crop pipeline. Pooled rows include relational questions. 95% bootstrap CIs; ✓ = CI clear of zero.

arm vs bar	V*Bench ($n=115$, $W=0.25$)		HR-Bench 4k ($n=400$, $W=0.15$)	
	Qwen3-VL-2B	Qwen2-VL-7B	Qwen3-VL-2B	Qwen2-VL-7B
DPR, single-object	+15.7 [+6.1, +25.2]✓	+11.3 [+1.7, +20.9]✓	+7.5 [+2.2, +12.5]✓	+8.5 [+3.8, +13.2]✓
argmax, single-object	+9.6 [+0.0, +19.1] [†]	+6.1 [-3.5, +15.7]	+2.5 [-2.8, +7.8]	-5.8 [-10.8, -0.8]
DPR, relational	-3.9 [-18.4, +10.5]	+0.0 [-13.2, +13.2]	-13.8 [-19.5, -7.8]	-8.8 [-13.8, -3.5]
DPR, pooled	+7.9 [-0.5, +16.2]	+6.8 [-1.0, +14.7]	-3.1 [-7.0, +0.9]	-0.1 [-3.6, +3.4]
bar (uniform@600)	62.6%	57.4%	64.5%	62.3%
DPR absolute	78.3%	68.7%	72.0%	70.8%

[†] best training-free read-out (max over a 4-layer window, App. F); the deployed mean-block argmax is lower.

3.1 Results at equal compute

Table 1 is the paper’s central result. On single-object questions the learned head clears the equal-compute bar in all four model×benchmark cells; the argmax feeding the identical crop pipeline clears in none. Against the unmodified model the gains are +15.0/ +14.1 pp; against random placement +28.3/ +26.7 pp. The HR-Bench 4k result is zero-shot — the head was trained on V*Bench and nothing was refitted — and holds under CircularEval (45.0% vs the bar’s 44.0% on Qwen2-VL; 44.0% vs 46.0% on Qwen3-VL, where the argmax is 40.0%).

Two things it does not do. It does not clear the bar pooled over question types on any cell, and it loses on relational questions on every cell. Both were predicted before the runs (§6) and both stay in every table.

3.2 Internal controls

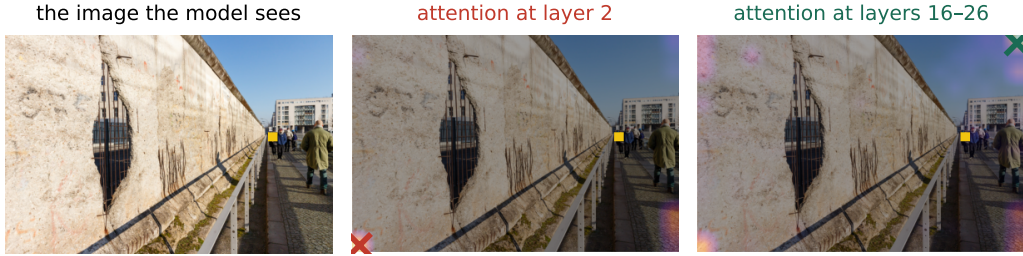
On the rows where the head and the argmax select the same cell (84 of 800 on HR-Bench 4k), the two arms are the same computation and their contrast is exactly +0.0 pp; a non-zero value there would void the headline. Re-running the head arm in a separate script reproduces the stored value on 100% of items on both models. The learned head’s gains are not geometry: a head given only position scores 3.7% coverage and a head trained on shuffled labels 2.6%, both at the rate of a random cell.

4 What the method is measured against

The deployed read-out — mean over layers 16–26, mean over heads, argmax — is the baseline this literature uses, and it is weak for four separable, nameable reasons. Reporting DPR only against it would overstate the head’s contribution by a factor of three (Table 2).

Axis 1: rule. CLAA (McDanel et al., 2026) diagnoses layer-wise ranking instability for LLM prefill with an answer-informed oracle and aggregates by max across a window of layers. Ported here: max over the deployed block is +7.3 [+2.6, +12.6] / +6.8 [+3.1, +11.0] pp of coverage on the two Qwen models; their 4-layer window-max, window chosen out-of-fold, +8.4/ +12.0. Mean over all layers and median are both worse than the block mean.

Axis 2: quantity. Kobayashi et al. (2020) showed in BERT that attention to special tokens is high- α , low- $\|v\|$. Our serialisation sink (App. D) is the same object: weighting each cell by $\alpha\|v\|$ is +4.7 [+2.1, +7.9] pp, and composes with the max to +10.5 [+5.8, +15.7] with no training and no selection.



"What is the color of the woman's scarf?"

Figure 2: The same forward pass, read at two depths. A real V*Bench item: attention over image cells at layer 2 (left) and in the late band (right), with the ground-truth box. The early map is not noisy — it is a confident map of something that is not the question. Reading it is worse than not reading anything (Table 3).

Axis 3: heads. Keeping the top 10% of heads by their visual-attention ratio (Lu et al., 2026) and taking the max is +11.5 [+5.8, +17.8] pp, label-free. CoRe’s (Tran et al., 2026) contrastive criterion, selected out-of-fold, is +3.1 [−1.0, +7.3] over it — not significant, and it needs boxes; the absolute criterion that fails in text works here.

Axis 4: prompt. The read-out is the final token’s attention. If the prompt does not end at the answer-emission point the map is unusable: with the answer options present but the instruction line removed, the argmax collapses to 3.7%/0.5% coverage — below the ~2% chance of hitting the target cell — while removing the options and keeping the instruction changes nothing (47.6/41.4 vs 46.1/39.8). This one line is why every HR-Bench 4k negative in an earlier version of this work was wrong (App. M), and it plausibly explains published “random beats attention” grounding nulls (Wen et al., 2025).

The fixes do not stack; the layer set does. Norm weighting, head selection and max each beat the deployed rule, but composed they land on the same ~56/~49%: they remove one sink by three routes. The component that adds on top is which layers: replacing the hand-set block with the layers where attention is question-conditioned — measured label-free by how much the map changes under four unrelated questions (§7) — gives 59.7/51.8%, within 3.7/2.7pp of the learned head (Table 2). The gate is threshold-robust (0.3–0.7 select the same layers, L17–20 on Qwen3-VL and L19–22 on Qwen2-VL) and independently re-finds Qwen2-VL’s answer-formation layer.

What does not transfer, and why that is a result. Attention rollout and gradient×attention — the standard faithfulness fixes in text — collapse to 12.6% and 15.7%, below the deployed rule. ReAttn’s entropy rescaling is catastrophic in both directions. Fixes that target the sink transfer; fixes that model information mixing destroy the signal. Rollout multiplies 28 attention matrices with residual identities and the query-conditioned spatial signal, which exists only in the late half, is washed out by the early layers it must pass through.

No training-free rule clears the bar on both models. End-task, the best label-free rule is +10.4 [+0.9, +20.9] pp over equal compute on Qwen3-VL single-object questions but +7.0 [−2.6, +16.5] on Qwen2-VL; the CLAA window-max is +9.6/ + 6.1. One of two, so under our rule it is not a bar-clearing method; it is the honest no-training counterpart, and the learned head leads it by 4–6 pp with lower bounds at or just under zero.

Table 2: The read-out ladder. Top-1 coverage of the evidence, V*Bench, $W=0.25$, ring-masked. Fixed constants throughout unless marked OOF. The learned head leads the best training-free rule by 3.7/2.7 pp, not the 17/15 pp it leads the deployed read-out by.

read-out	needs	Qwen3-VL-2B	Qwen2-VL-7B
deployed: mean over block, argmax	—	46.1	39.8
max over block (CLAA)	—	53.4	46.6
$\alpha\ v\ $, mean over block	$\ v\ $	50.8	42.9
$\alpha\ v\ $, max over block	$\ v\ $	56.5	48.7
top-10% heads by visual ratio, max	per-head	55.0	48.7
max over a 4-layer window (OOF)	boxes	55.0	51.3
CoRe heads (OOF), max	boxes	58.1	—
max over question-conditioned layers	label-free	56.0	52.4
+ heads + $\ v\ $	label-free	59.7	51.8
DPR learned head (OOF)	boxes	63.4	54.5
readable head, $W=0.15$ basis (§4.2)	boxes	parity at $W=0.15$	
attention rollout (Abnar & Zuidema, 2020)	—	12.6	—
gradient $\times$ attention (Chefer et al., 2021)	—	15.7	—
ReAttn entropy rescaling (Tian et al., 2026)	—	−19 to −27 pp	

Table 3: Pruning 90% of visual tokens from layer 2, ranked at different depths. Late = mean over the deployed block. Layer-2 is below random selection in all six cells; late beats layer-2 with a clear CI in all six. “Late is free” holds on V*Bench only.

benchmark	model	none	random	layer-2	late	late−layer-2	layer-2−random
V*Bench	Qwen3-VL-2B	56.5	38.2	34.6	56.0	+21.4	−3.7
V*Bench	Qwen2-VL-7B	52.9	44.5	39.8	51.3	+11.5 [+4.2, +18.8]	−4.7
POPE	Qwen3-VL-2B	90.5	82.5	67.5	84.0	+16.5	−15.0
POPE	Qwen2-VL-7B	88.0	83.0	72.0	84.0	+12.0 [+5.0, +19.0]	−11.0 [−17.0, −5.0]
MMBench	Qwen3-VL-2B	85.5	78.5	53.0	80.0	+27.0	−25.5
MMBench	Qwen2-VL-7B	85.5	80.5	75.0	83.0	+8.0 [+3.5, +12.5]	−5.5 [−10.5, −1.0]

4.1 Ranking visual tokens by early attention is below random

The same read-out is the ranking signal in attention-based token pruning, where the widely used default reads layer 2 (Chen et al., 2024). We pin the prune point at layer 2 for every arm — identical budget, identical savings — and move only the layer the ranking is read from. This decomposition is, to our knowledge, unrun: every surveyed method moves read depth and prune depth together (§8).

Table 3 is the paper’s most widely replicated result: two models, three benchmarks, and the effect is larger on general VQA than on the small-object benchmark that might flatter it. It reconciles two published results that appear to contradict: Wang et al. (2026) find attention-based selection no better than random when read and pruned deep; we find a deep read beats random by 17.8 pp when pruned early. Both hold, because read depth and prune depth are two knobs. A pre-registered mechanism prediction failed here and is reported as failed: Qwen2-VL’s early layers rank the target worse (gt-pct 0.620 vs 0.456) yet its layer-2 penalty is smaller (−13.1 vs −21.9 pp); ranking quality does not predict pruning damage.

4.2 A readable head at parity

The gradient-boosted tree is accurate but opaque. Four rounds of replacement — an MLP, cross-attention over the layer axis, a spatial CNN over the 28-channel map with MSE and pairwise losses, additive GAMs, rank fusion, ensembles — fail to beat it on both Qwen models (App. E); the constraint is 191 boxed items, not model class. What does reach parity is a Lasso-fit log-linear model with ~ 91 parameters: $\sum_l w_l \log A_l + \sum_l v_l \text{rank}_l + \sum_l u_l \log \bar{A}_l^{3 \times 3} + \text{geometry}$. At the deployed $W=0.15$ it is +2.6/ +1.6 pp over the tree on

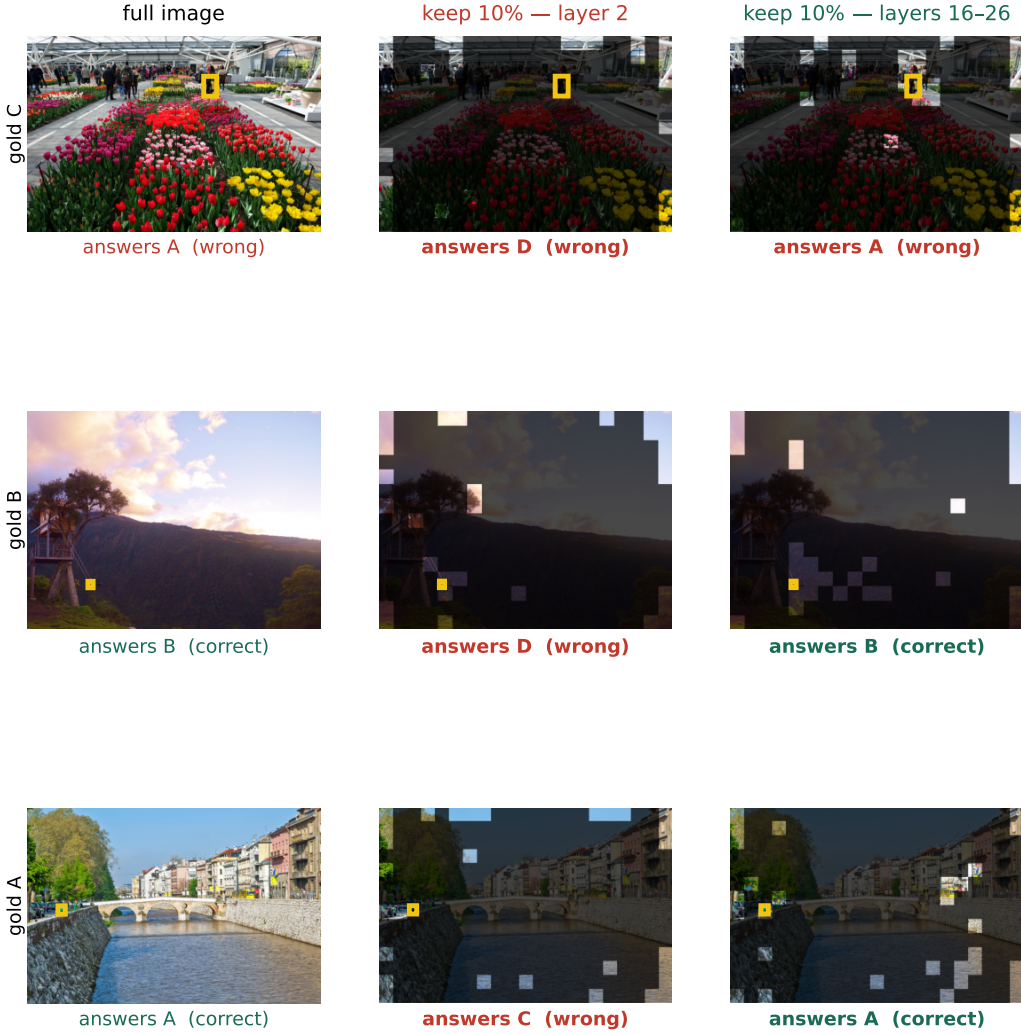


Figure 3: What each ranking actually keeps. Unshaded cells are the 10% of visual tokens retained under a layer-2 ranking (left) and a late ranking (right) on real items, with the evidence box. The layer-2 set is dense at the image boundary — the sink — and misses the target.

Qwen3/Qwen2 (CIs span zero), +4.7 on LLaVA-NeXT and +6.3 [+1.6, +11.5] on LLaVA-OneVision — the one model the tree fails on — and it clears the deployed argmax on four of four architectures where the tree manages three. Its weights say what the tree learned: L19 strongest on both models, L21 on Qwen2-VL (its answer-formation layer, §5), the last layers subtracted on Qwen3-VL, neighbourhood terms carrying $1.5\text{--}2.3\times$ the mass of point terms, sink indicators ≈ 0 . Rank fusion alone loses 25 pp: the magnitudes carry the signal, which is also why norm weighting helps and rollout fails. We use the tree for the numbers in Table 1 and the readable head to say what they mean.

5 Why it works: the evidence is not encoded

The cliff. Define tokens-on-target as the target’s area fraction times the visual budget. Below 0.25, uniform encoding scores 32.4% [21, 44] and 36.6% [25, 48] on the two models — on the cleanest single-region subset, 19.0%, under the 25% chance level — while the oracle crop at the same 300 tokens scores 95.8% on both. Above it, 77.3/63.6% against an

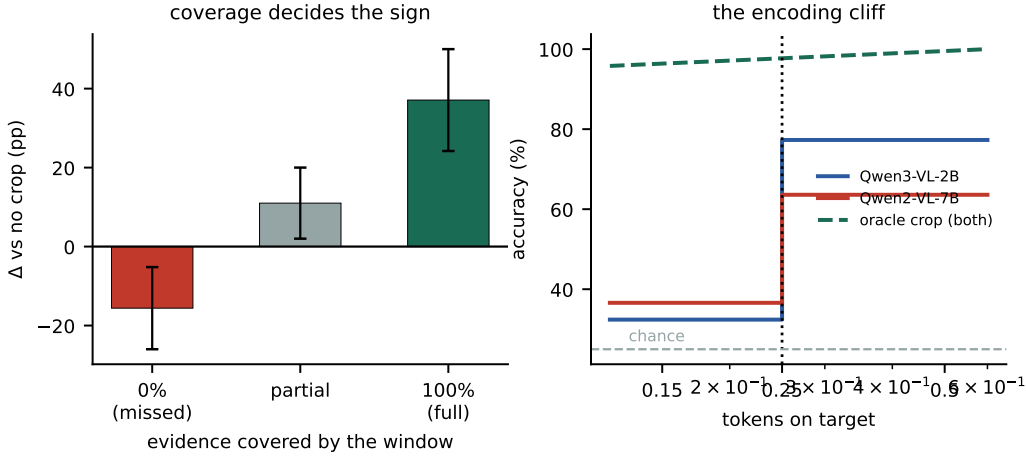


Figure 4: Left: coverage of the evidence set by the crop window decides the sign of the allocation effect, crossing zero near 25%; half of all windows miss. Right: the encoding cliff. Uniform encoding falls below the 25% chance line under ≈ 0.25 merged tokens of target extent while the oracle crop is flat at 96–100% across it, on both models.

oracle at 100%. The oracle arm is the control: it is flat across the split, so the items below the threshold are not harder, they are invisible. Stated as Li et al. (2026)’s RT_{50} in our units, the threshold is ≈ 0.15 – 0.25 merged tokens on both checkpoints; their synthetic-canvas threshold for Qwen3-VL-2B converts to ≈ 0.14 – 0.39 .

Nine in ten errors are encoding failures. Of every V*Bench error, the oracle crop fixes 86.7% (Qwen3-VL) and 84.0% (Qwen2-VL) at $W=0.25$, and 94.0/89.4% at $W=0.15$. Same model, same question, same budget; only the placement differs.

Seven interventions fail, two of them decisively. Activation steering, sink suppression, residual-stream injection, DoLa/DeCo (Chuang et al., 2024), component ablation, attention amplification and contrastive decoding are all null (App. J). The two that close the argument: amplifying attention on exactly the right cells recovers +5.8 pp against the crop’s +36.2 pp at the same budget — 16% — because the oracle amplification set is one cell of 294; there is nothing to amplify. And the region is causally live: masking it moves the output $3.32\times$ more than a random region and costs 10.5 pp, yet contrastive decoding steered to it has a ceiling of +2.6 [−0.5, +5.8]. The model looks in the right place; the right place is load-bearing; it still cannot answer, because no operation on the residual stream supplies information the encoder never wrote.

What the crop restores. Read the answer at every layer (Fig. 5). The cropped and uncropped arms are indistinguishable through layer 20, then separate abruptly. Split the same method by whether its window actually contained the evidence: +29.7 [+19.8, +39.6] where it covered and −7.9 [−18.0, +2.2] where it missed on Qwen3-VL; +23.8 [+13.1, +34.5] and +0.0 [−11.2, +11.2] on Qwen2-VL. Same computation, same budget, same question; only coverage differs. Coverage is the mediator and it decides the sign (Fig. 4): a missed window costs −15.6 pp, a fully covering one gains +37.1 pp, and within a single difficulty stratum the gap is 70.8 pp with missed windows at 13.0%, below chance. The method pays exactly where the cliff says it should: +13.8 pp below 0.15 tokens, +16.0 in the cliff zone, +3.7 (n.s.) above.

Against the predecessor’s reading. Liu et al. (2026) conclude from the same kind of attention maps that VLMs see but do not believe: they attend to evidence they fail to use. That holds above the threshold. Below it, the correct option is decodable at no layer on items the model localises but answers wrongly (72.2% ever-argmax vs 68.1% for items whose window

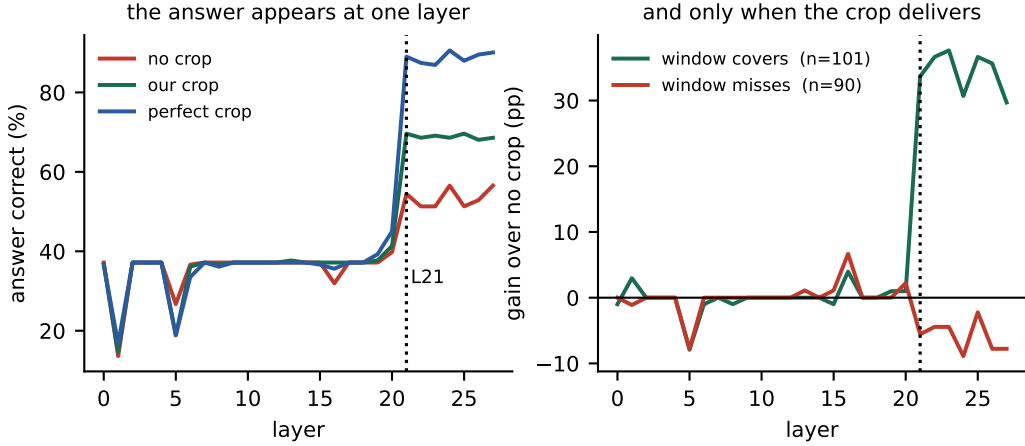


Figure 5: Cropping restores an answer-formation step; it does not improve reasoning. Logit-lens accuracy at every layer for four arms that differ only in the pixels shown. The arms are indistinguishable for twenty layers, then separate by 15 pp at layer 21. The failing arms jump at layer 2 — a prior forming — and never improve: their maximum over all 28 layers equals their final answer. Final-layer values come from `model.logits`, never a re-normalised hidden state (App. A).

missed entirely — no difference) and the oracle crop fixes 94.4% of them. The model points but never saw. The two accounts divide at the cliff, and neither VEA, Lu et al. (2026) nor Esmaeilkhani & Latecki (2026) measures target size or runs the oracle control that locates it; the last is the honest exception, and it works by changing the weights — its own ablation shows target-token norms rising 6–19%.

6 Where it stops

Every limit below was found by a control we ran, and each has a mechanism.

Relational questions. DPR loses on them in every cell (Table 1). The evidence set of a relational question is the union of the objects involved, $7.9\times$ larger in area than a single-object target; one window cannot cover it, and a perfect crop of one region wins (+19.7 pp), so it is coverage and not question difficulty. Predicted from the coverage account before any run. This is the first claim in the project to replicate across two models and two benchmarks, and it is a limit.

Existence questions. On POPE the crop arm is -56.7 , -77.3 and -63.0 pp against the bar on the three positive strata, because it answers “no” to almost everything: for “is there an X in the image?” a window showing 6% of the area is genuine evidence of absence. The attention localiser is at chance there (9.2% centre-in-box vs 10.7% expected) — existence questions do not demand localisation — and the oracle arm is circular: knowing the box is knowing the object is present. So POPE cannot supply a ceiling, and crop-based allocation is invalid, not merely unhelpful, for questions that do not presuppose their target. The pre-registered prediction that cropping would raise false positives was wrong in direction (they fell 4.0 pp), for the same reason.

One architecture whose layers agree. On LLaVA-OneVision the learned head is -1.0 [-6.3 , $+3.7$] and max-over-layers is -3.7 : it is the one model where the best single layer equals the block mean, so there is no disagreement across depth for any re-ranker to exploit. The head is 3 of 4 architectures on proposals ($+14.1$, $+8.9$, $+7.3$ [$+2.1$, $+12.6$]); the readable head of §4.2 is 4 of 4.

The read-out is task-specific. The obvious lever against a 191-item ceiling is more boxed data. We trained the head on 3,000 TextVQA items with public evidence boxes and applied it zero-shot to V*Bench: 38.2% coverage at $W=0.15$ and 37.2% at $W=0.25$, **below the deployed argmax** (39.3/46.6), and mixing them into the V*Bench folds degrades the head ($-5.8 [-11.0, -1.0]$). The same head beats the argmax on TextVQA itself (46.5 vs 38.3), so it learns; what it learns does not carry. The reason is visible in the layer profiles: question-conditioned localisation switches on at L11–13 for text reading and at L16 for small-object attributes, so a head trained on the former trusts layers that are still question-blind on the latter. Transfer works within a question type (V*Bench $\rightarrow$ HR-Bench 4k) and fails across — the VLM counterpart of Orgad et al. (2025)’s “skill-specific” probes, and of Lu et al. (2026)’s task-dependent boundary bands.

7 Mechanism

Attention is question-blind until half depth, on four architectures. Run the same image under four unrelated questions and measure per-layer divergence of the attention map — no labels. It is 0.001–0.005 through L12 on both Qwen models, then jumps $13\times$ by L16 (0.137/0.178); the switch-on is at L14/32 (44% depth) on LLaVA-NeXT and L16/28 (57%) on LLaVA-OneVision. This is the band CLAA, Lu et al. (2026) (their transition zone, 37–63% depth) and VEA (L16–26) independently identify — four measurements, two modalities. The locator finds the region, not the best layer: on Qwen2-VL divergence rises at L13–14 but pruning damage recovers at L14–16, and reading at L14 still loses 6.3 pp to L16 (App. C).

Localisation is built by the language model. Every read of the vision tower’s 24 layers is at or below the 2.3% chance rate (best layer 1.6%, learned combination 1.0%, mean 0.5%) while the language model reaches 39.3% on the same items. The encoder never sees the question. Attention from the noun naming the target is near chance (gt-pct 0.367 vs 0.500): localisation is not lexical lookup and exists only after the whole query is integrated, at the answer position (§4).

Attention quality and answer formation are dissociated. Attention localises best at L17–19; the answer forms at L21; rank correlation across layers $\rho=0.297$. Attention top-1 is higher in the six layers before the answer forms (30.1%) than after (24.8%). Raised by an external reviewer, tested, confirmed: these are two quantities, not one.

The serialisation sink. The image is flattened row by row; the sink sits at the row boundaries. Last column $5.1\times$ its share of attention, first column $2.9\times$, rows $0.9/0.8\times$ — columnar, not cornered. A transpose test (14×21 vs 21×14) keeps the sink at the same (row, col) positions with only $4/8$ shared absolute indices, so these are not register tokens; replaying sink indices into the interior lands below chance (1.1% vs 3.3%). The prediction only this account makes: models that splice a learned image_newline at each row end put the sink on the separator ($2.0\text{--}2.3\times$), which is not a corner, not a pixel and not image content. Four architectures, two families (App. D). Suppressing it buys nothing; biasing any 4.8% of image tokens moves accuracy by ≤ 2 pp while the same number of text tokens costs 17 pp.

8 Related work

Attention reweighting — VEA (Liu et al., 2026), head rescaling (Lu et al., 2026), KLAL (Esmaeilkhani & Latecki, 2026) — reports gains, measures no target size and runs no oracle control; §5 locates the family’s boundary. VEA finds L16–26 by AUROC on document VQA; we find the same band label-free on small-object search, a convergence we cite rather than claim. Cropping and allocation — ViCrop (Zhang et al., 2025), ZoomEye (Shen et al., 2024), AttWarp (Anonymous, 2026a), Visual-CoT (Shao et al., 2024), CropVLM (Carvalho et al., 2026) — is scored at matched realised budget in App. G: every policy breaks even or loses (ZoomEye to a single uniform image of its own budget; grounding -6.2 pp at median IoU 0.029; AttWarp $+0.6$; tree search at 4K -15.3 at $2.2\times$ the tokens, $n=800$), and so

does our own fixed always-crop policy; only the learned read-out passes. Token pruning — FastV (Chen et al., 2024), FEATHER (Endo et al., 2025), TwigVLM (Shao et al., 2025), LearnPruner (Anonymous, 2026b), Wen et al. (2025), Wang et al. (2026) — establishes that early-layer attention is unreliable; none runs the read-depth/prune-depth decomposition of §4.1 or reports the four axes of §4. Interpretability transfer — rollout (Abnar & Zuidema, 2020), norms (Kobayashi et al., 2020), LRP (Chefer et al., 2021), CLAA (McDanel et al., 2026), CoRe (Tran et al., 2026), ReAttn (Tian et al., 2026), Orgad et al. (2025) — is ported and scored in §4 and App. K. Fine-scale perception: Li et al. (2026) find perception saturating near 12px on synthetic canvases; the cliff is its natural-image, token-unit, oracle-controlled counterpart.

9 What does not work

Reported at equal weight because it is most of what we learned. Each entry has a pre-registered rule applied as written; details in App. L.

- Per-item window size. Pre-registered, dead on two models and two grids: -3.1 pp against the fixed constant it was meant to improve.
- Gates. The two-pass confidence rule that rescued the old proposer is null for the head (-0.5); a free routing gate clears on one model only and was barred by pre-registration from substituting for the sizer.
- Multi-crop. Four ranked windows in one pass: predicted $+5.5$, measured -3.1 ; the coverage model has no term for distractor cost.
- More capacity. MLP, cross-attention over layers, spatial CNN, GAM, ensembles: nothing beats the tree on both models; a 91-parameter model matches it.
- More features. Giving the head the max-aggregate and $\alpha\|v\|$ signals that rescue the simple read-out adds nothing: it already extracts them.
- More data. 3,000 cross-task boxed items train a head worse than no head.
- Pre-generation detection. The attention depth profile does not detect certified encoding failures (1 of 2, ordering reversed); plain predictive entropy is the best detector on both models.
- A token-level cliff. Evidence tokens were at chance below the cliff on one model (0.513 vs 0.656 above) and not on the other (0.623 vs 0.633); withdrawn.
- Two of our own diagnostics, refuted by the measurements that were meant to confirm them.

10 Limitations

The end-task win is single-object questions on two Qwen checkpoints and two benchmarks; pooled it does not clear anywhere. LLaVA is tested on proposals only. The cliff, the interventions and the mechanism are V*Bench-only because it is the only venue with boxes where the oracle control is valid — POPE’s oracle is circular, HR-Bench 4k has no boxes — and five of the seven intervention nulls are single-model. The head is task-specific and needs boxes on its training benchmark. The equal-compute bar counts tokens, not latency; two 300-token passes and one 600-token pass are not identical in wall-clock. Every number in this paper was obtained with $n=191$ or $n=800$ items and a 1–2.5 pp run-to-run floor on coverage; lower bounds of $+1.7$ and $+2.2$ are real but thin.

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Appendix contents

A Protocol · B Replication ledger · C Depth structure · D The sink · E The head and its replacements · F The training-free ladder · G Prior art at matched budget · H 4K transfer · I POPE · J Seven interventions · K Probes · L Negatives in full · M Corrections · N Reproducibility.

A Experimental protocol

Budgets are measured, never computed. Visual-token counts are read from `image_grid_thw` after processing; images are iteratively resized to hit a target (tolerance 6%). A contrast whose arms drift by $> 10\%$ is void. Qwen3-VL’s processor floors every image at 64 merged tokens, so tiling arms below 5×64 are infeasible rather than mismatched.

Localisation pass. The full prompt — question, options where they exist, and the answer instruction — with the image at 300 tokens; last-token attention over image cells, head-averaged, each layer sum-normalised. Border cells ring-masked before any argmax. LLaVA models: `image_newline` separators are detected by matching the model’s own embedding against the merged sequence, stripped, and the grid rebuilt from the modal row length; a naive split once swallowed the separator-free base image as “row 0” and put 59.8% of positions in a bucket that can hold $\sim 5\%$.

Answer read-out. Softmax over the option letters’ logits at the final position; yes/no over the Yes/No token families. No generation, so results are deterministic.

The logit lens. Intermediate layers: `lm_head(final_norm(h_i))`. The final layer must be read from `model.logits`: HuggingFace returns `hidden_states[-1]` already normalised, and normalising it again corrupted only the last layer, manufacturing a +5.1 pp “free read-out gain” and three sections we retracted (App. M); the two implementations agreed on L24 for 100% of items and on the final layer for 82.2%, which is how it was found.

Cross-validation. `GroupKFold(5)` grouped by item, three fold-permutation seeds, training rows negative-subsampled to every covering cell plus 30 random non-covering cells per item; evaluation on all cells. In-sample layer selection reads 41.9% on Qwen3-VL and 36.1% out-of-fold — below the block mean it appeared to beat — which is why nothing here is selected in-sample.

Statistics and the noise floor. Item-level bootstrap, 4000–8000 resamples. The same head on the same items under two attention extractions differs by ~ 1 pp; under two fold seedings by up to 2.5 pp. No coverage difference inside that band is reported as signal.

Numerical precision. Qwen2-VL loaded in fp16 produces all-NaN logits; argmax falls through to index 0, and because 71/191 V*Bench labels are “A” every arm scores an identical 37.2% with CI $[+0.0, +0.0]$. Two false negatives that looked exactly like findings; bf16 throughout, and an arm contrast of exactly $+0.0 [+0.0, +0.0]$ is treated as a pipeline fault, never a null.

B Replication ledger

C Depth structure across architectures

Averaging. Mean over all 28 layers is worse than the deployed block on all four architectures (36.6/21.5/20.4/9.9% vs 39.3/35.1/25.1/12.6 at $W=0.15$); the learned read-out is $+6.3 [+2.6, +10.5]$, $+8.4 [+3.1, +13.6]$, $+1.0 [-2.6, +5.2]$ (null) and $+6.3 [+2.1, +10.5]$. The remedy’s form is architecture-dependent: Qwen2-VL is fixed by one integer (L21, stable in all five folds, +8.4 pp); on Qwen3-VL no single layer is stable out-of-fold and a learned combination is needed.

Table 4: Every claim by its replication level. Q3 = Qwen3-VL-2B, Q2 = Qwen2-VL-7B, NX = LLaVA-NeXT-7B, OV = LLaVA-OneVision-7B. V = V*Bench, H = HR-Bench 4k, P = POPE, M = MMBench.

claim	models	benchmarks
DPR beats equal compute, single-object	Q3, Q2	V, H
Layer-2 pruning below random	Q3, Q2	V, P, M
Cropping does not beat the bar on relational questions	Q3, Q2	V, H
Attention question-blind until ~half depth	Q3, Q2, NX, OV	V
Serialisation sink at row boundaries	Q3, Q2, NX, OV	V
Averaging across depth dilutes the read-out	Q3, Q2, NX	V
Max beats mean as the layer rule	Q3, Q2	V
Encoding cliff, oracle flat across it	Q3, Q2	V
87/84% of errors oracle-fixable	Q3, Q2	V
Answer formation restored at L21, only where the crop delivers	Q3, Q2	V
Coverage decides the sign	Q3, Q2	V
The read-out needs the answer-emission point	Q3, Q2	V
Orgad's exact-answer-token probing does not port	Q3, Q2	V
Readable head at parity with the tree	Q3, Q2 (parity); NX, OV (ahead)	V
Norm weighting; head selection; label-free layer gate	Q3 (Q2 for gate)	V
Vision tower at chance; noun-token attention at chance	Q3	V
Amplification 16%; region load-bearing 3.32×	Q3	V
Attention quality / answer formation dissociated	Q3	V

Rejected by replication: final layer anti-correlated (Q3 only); signed contrast is the mechanism (failed twice); no single layer is a good localiser (false on Q2); rank-only multi-crop; per-item sizer (2 models × 2 grids); depth-profile detector; token-level cliff; head beats CLAA on coverage (1 of 2); coverage interaction on HR-Bench 4k (1 of 2); training-free rule clears the bar (1 of 2).

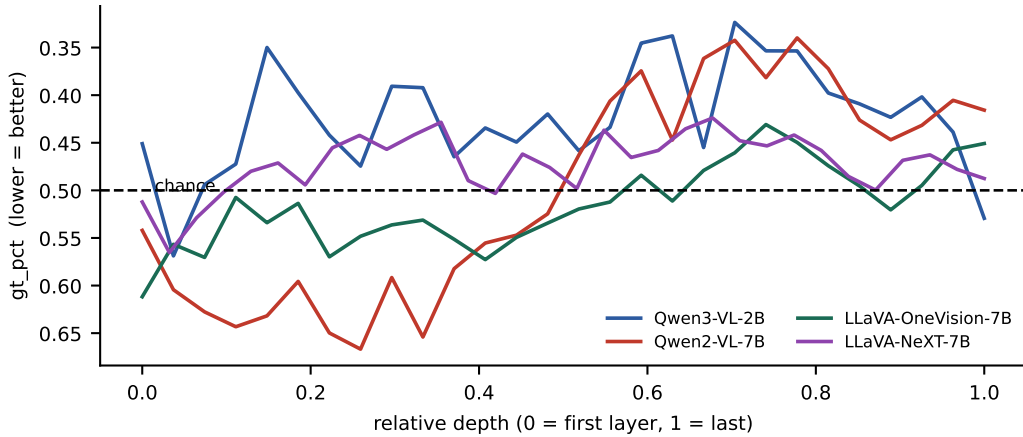


Figure 6: Per-layer localisation quality against relative depth, four architectures. 0.500 is chance (lower is better). Which layers are anti-correlated with the target differs by architecture: the final layer is at 0.529 on Qwen3-VL and 0.416–0.488 elsewhere, which is why “the final layer is anti-correlated” was rejected as a general claim.

Read-depth sweep on Qwen2-VL. Pruning at layer 2, ranked at each read depth: L0–L12 36.6–40.8% (flat floor), L14 44.0 (+4.7 [−1.0, +10.5] vs L2, not clear), L16 50.3 (+11.0 [+3.7, +18.8]), L20–L26 48.7–50.3 (L20 – L16 = −1.0 [−6.3, +4.2]). One transition, then a plateau: layer 2 is not anomalous, the whole first half is equally bad.

Question divergence. Same image, four unrelated questions, mean L_1 distance between per-layer maps: Qwen3-VL 0.001–0.005 (L0–L12), 0.0105 (L14), 0.1374 (L16); Qwen2-VL 0.0046 (L12), 0.0130 (L13), 0.0442 (L14), 0.1779 (L16). LLaVA-NeXT switches on at L14/32, LLaVA-OneVision at L16/28. A detection rule set at 3× baseline fires on noise (it flagged L5 at 0.0055); real signal is 10–100×.

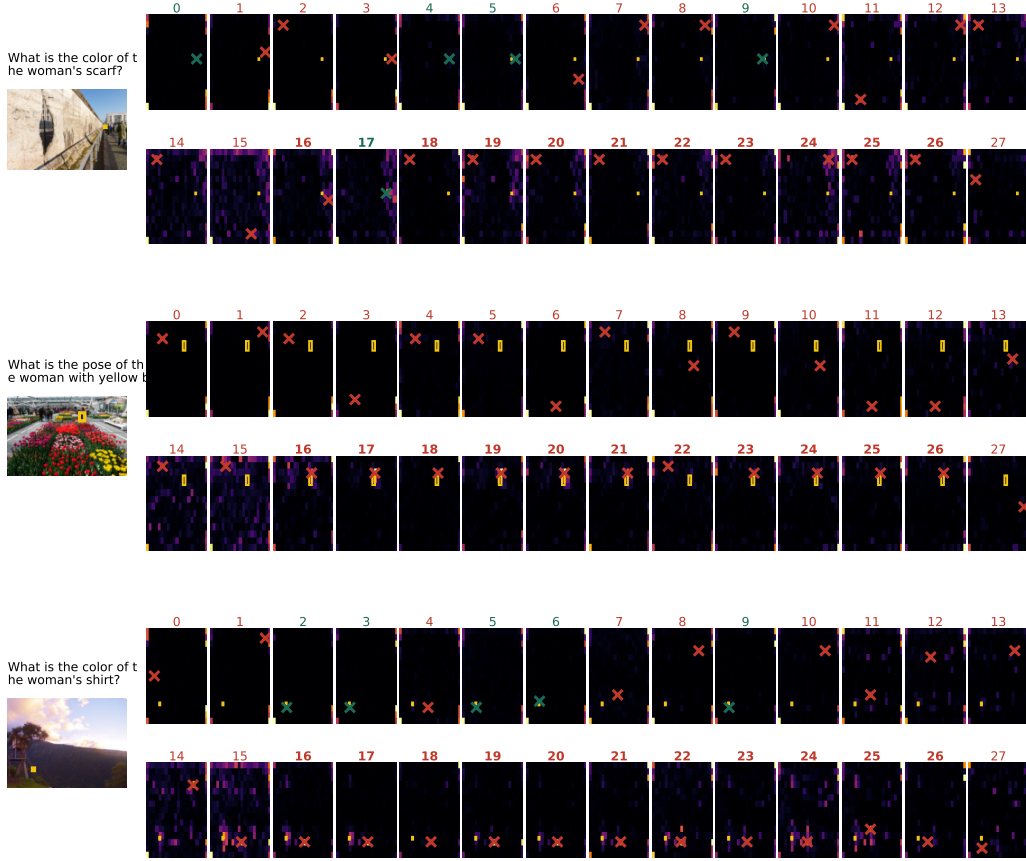


Figure 7: Every layer, three items, one forward pass each. Attention over image positions at all 28 layers with the ground-truth box; layers whose top-1 cell would cover the target under $W=0.15$ are marked. Covering layers are scattered and mostly outside the averaged block (L0, 4, 5, 9, 17 and L2, 3, 5, 6, 9 on two items), which is what a learned read-out over the full profile exploits and a block mean cannot.

Cut blind early, rank late. Attention-guided and random early cuts to 50% are indistinguishable (51.8 vs 50.8% at 10% final keep; +1.0pp, CI spanning zero), which is the question-blindness claim applied to the cut itself; the two-stage saves 64% of visual-token compute against pure-late’s 39% at $-4.7 [-9.9, +0.5]$. Prior art: STAR and LearnPruner are two-stage; this is reported as a measurement of the blind-cut tolerance, not as a method.

D The serialisation sink

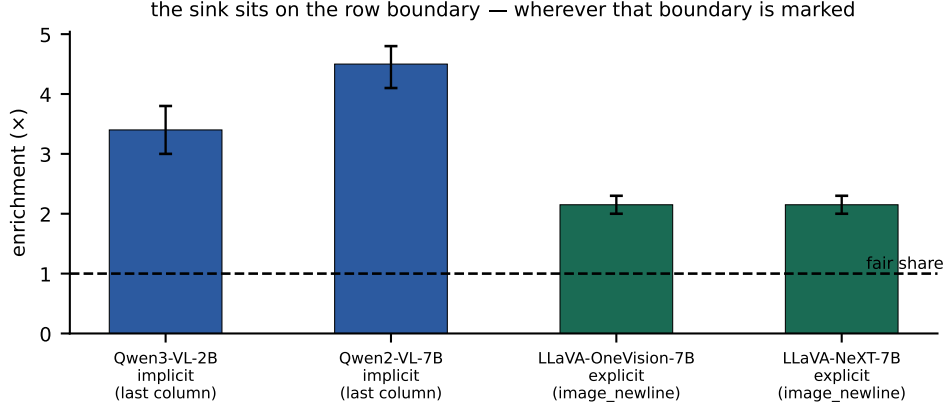


Figure 8: Trailing-boundary enrichment on four architectures. Where the row boundary is implicit (Qwen) the sink is the last column, 3.4–4.5 $\times$; where a learned image_newline is spliced at each row end (LLaVA) the sink sits on the separator, 2.0–2.3 $\times$. A correction on the record: LLaVA’s bottom rows are also enriched 2.1–3.0 $\times$, so “rows carry no excess mass” is Qwen-specific.

The sinks absorb 41–45% of the selected attention mass on the deployed read-out; the ground-truth cell ranks in the top 3.4% of cells while the top-1 cell is almost never the target. The learned head’s ablation attributes its gain to the depth profile (+7.3pp) and none to the sink indicators (+0.0): it does not fix the sink, it routes around it. Suppression at inference (phase 50) is indistinguishable from suppressing random or interior tokens.

E The learned head and its replacements

Ablation (Qwen3-VL, coverage). Full head 52.9% ($W=0.15$); without the depth profile -7.3 ; without sink indicators -0.0 ; geometry-only 3.7; shuffled labels 2.6; attention features alone deliver +11.5 of the +13.6.

head, $W=0.15$	Q3	Q2	NX	OV
deployed argmax	39.3	35.1	12.6	25.1
GBT (incumbent)	53.4	44.0	19.9	24.1
MLP, MSE (ph. 102)	49.2	47.1*	—	—
cross-attention over layers, listwise (ph. 102)	51.8	43.5	—	—
spatial CNN 3 $\times$ 3, MSE (ph. 130)	$-6.8 [-12.6, -1.0]$	$-8.9 [-15.7, -2.1]$	+1.0	-2.6
spatial CNN, pairwise	-1.0	-4.7	+0.0	-1.6
GBT+CNN rank ensemble	+0.0	-2.1	+1.6	+3.1
log-linear, 63 params (ph. 131)	-2.6	-2.1	+1.0	+6.8 [+2.1, +11.5]
additive GAM	-8.4	+0.0	-1.6	+4.2
rank fusion	-27.2	-25.1	-4.7	-11.0
log-linear + per-layer 3 $\times$ 3 means, 91 params (ph. 132)	+2.6	+1.6	+4.7	+6.3 [+1.6, +11.5]

Rows below the incumbent are Δ vs GBT. *phase 102’s incumbent was 47.1 under its own seeding.

Architecture search (four rounds, all OOF on identical folds). Adding the max-aggregate and $\alpha||v||$ signals as features: +0.0, -0.5 , +1.6 $[-1.0, +4.2]$ (ph. 112). Up-weighting

encoding-failure items: monotonically worse, $-6.8 [-11.5, -2.6]$ at $w=10$ on Q2 (ph.118). Alternative targets (classification, multi- W , squared coverage): none beats regressing coverage@0.25 (ph.122). Cross-model transfer of the tree: Q3-trained on Q2 maps 46.6 (vs its own 54.5), Q2-trained on Q3 maps 55.5 (vs 63.4).

The parity head’s weights ($W=0.15$, per-layer point + neighbourhood). Qwen3-VL: L19 +.112, L5, L17, L24, L8 positive; L10 -0.101 , L26 -0.096 , L3, L22, L23 negative; neighbourhood mass $2.30\times$ point mass. Qwen2-VL: L19 +.079, L21 +.071, L16 positive; L17, L7, L11, L20 negative; $1.46\times$. Sink flags $|w| < 0.02$ on both.

F The training-free ladder

All fixed constants unless marked. Block L16–26 (Q3) / L15–26 (Q2); $q=0.10$; divergence gate $\geq 0.5 \cdot \max$, robust across 0.3–0.7. Components tested and rejected: ReAttn entropy rescaling (-19 to -27 pp on norm-weighted maps, -2 to -8 on raw — early layers are diffuse and any entropy reweighting promotes them); VEA’s 3×3 outlier denoising at $\lambda=10$ (-1 to -3 pp on Qwen; isolated high-value cells on V*Bench are usually the target, the opposite of document VQA); leave-one-out background normalisation (null-to-negative once max/heads are present). End-task at $W=0.25$, single-object vs uniform@600, head arm re-run as control (100% agreement): learned head $+15.7/ +11.3$; label-free gated max $+10.4 [+0.9, +20.9] / +7.0 [-2.6, +16.5]$; composite $+9.6/ +7.8$; CLAA window-max $+9.6/ +6.1$. Norm-weighted max end-task on Qwen3-VL (ph.113): 68.6% pooled, not clearing. Cross-family: no training-free rule helps LLaVA; max hurts OneVision (-4.2); the divergence mechanism replicates there but gating by it does not rescue localisation.

G Prior art at matched budget

V*Bench, Qwen3-VL-2B, every arm scored against the uniform sweep interpolated at its own realised token count. ZoomEye (Shen et al., 2024): highest raw accuracy in the table, loses to one uniform image of its own budget. Grounding (“locate the X and output its box”): -6.2 pp, boxes parse on 98% of items with median IoU 0.029, because it must ground on the same downsampled view that cannot resolve the target. AttWarp (Anonymous, 2026a): same attention source, same inverse-CDF warp, scored against the bar its paper never runs: $+0.6$ pp, break-even, and it costs two passes. Our fixed always-crop policy: 60.2% against a 66.0% bar, captures 10% of the oracle gain; a conditional allocator answering from the more confident pass clears it (68.6%), and its learned gates lost to a rule. At 4K ($n=800$): tree search -15.3 pp against its own bar and -10.4 against uniform@1200 at $2.2\times$ the tokens, worst arm per token; grounding flips sign across scales ($-6.2 \rightarrow +1.3$) exactly as the coverage account predicts. Gating recovers $+5.7$ pp for ZoomEye while cutting its tokens 53%: tree search wastes most of its nine passes on items no crop helps. The exchange rate: no uniform rung reaches the crop arm on V*Bench up to the model’s ceiling, a lower bound of $\geq 26\times$ set by the architecture; on LLaVA-NeXT there is no budget axis to move along at all.

H Zero-shot transfer to HR-Bench 4k

Head trained on all 191 V*Bench items, applied with nothing refitted — not W , not the block, not the mask. Instance keying includes the image hash: 159 unique question strings across 200 instances meant keying on (question, category) once paired 212/800 rows with the wrong image. Proposals: $+4.9$ pp coverage over the argmax, $+6.5$ CircularEval, $+8.5$ single-region. End task in Table 1. Qwen2-VL head trained separately on its own V*Bench maps (block L15–26). The W sweep at 4K ($W \in \{0.06, \dots, 0.25\}$, argmax proposer): single-object flat (69.9–71.1%), relational monotone in W ($37.3 \rightarrow 55.2$), and the interaction $W_{0.12} - W_{0.25}$ is $-11.9 [-23.9, -1.5]$ on relational for Qwen3-VL but $-3.0 [-11.9, +6.0]$ on Qwen2-VL — one of two, so the coverage interaction is claimed on V*Bench only while the relational boundary holds on both.

I POPE as a scope test

RePOPE-clean labels joined to COCO boxes, stratified to over-represent the below-cliff band (90 positives < 0.25 tokens, all taken; 198 in $[0.25, 1)$; 200 above; 200 negatives), so overall accuracy is not comparable to published POPE. Budget drift 0.1%. The median

stratum	n	uniform@300	uniform@600	crop@0.25	random crop	oracle crop
below cliff	90	76.7	70.0	13.3	6.7	93.3°
cliff zone	198	84.8	89.4	12.1	16.7	97.0°
above	200	100.0	98.5	35.5	44.5	89.5°
negatives	200	88.5	89.5	93.5	94.0	—

°circular: on POPE, knowing the box is knowing the object is present.

POPE object occupies 32 merged tokens at $B=300$ and only 90 positives fall below the cliff; 27.7% of the sample has a box larger than a $W=0.25$ window, so the coverage metric is undefined there and the size-independent centre-in-box metric is used: deployed argmax 9.2% [6.8, 11.9] against 10.7% expected. (The “0.4 merged tokens” figure elsewhere in this work is POPE’s confident-denial cohort, a selected subset.) H-POPE (Pham & Schott, 2024) fixes the question form but uses the same 640×480 COCO images: right question, no magnification headroom.

J Seven interventions

All on Qwen3-VL unless stated; one pass at identical tokens, so no bar to clear. (1) Activation steering at L16, direction fit on disjoint items: null; a `rand_last` control raised $P(\text{yes})$ by shifting bias, after which every arm reports a false-positive rate. (2) Sink suppression: indistinguishable from suppressing random or interior tokens; suppressing the same number of prompt tokens costs 17.3 pp. (3) Attention amplification on the oracle cells: +5.8 vs the crop’s +36.2 at the same budget; every practical arm (top-1/5/15, deployed window) indistinguishable from random. (4) Residual-stream injection: a shared “allocate to evidence” direction exists (cosine 0.53–0.75) and is answer-irrelevant (+0.0 pp at every layer for the mean direction; an item’s own displacement at L20 recovers 47%; L26 recovers everything and is near-tautological). (5) DoLa dynamic +0.0, DeCo best-of-8 −0.7, read-at-L24 +0.0, with their hyper-parameters swept in their favour. (6) Zeroing attention or MLP in L25–27: ≤ 2.4 pp, never significant. (7) Contrastive decoding steered to the predicted region +2.1, to the ground-truth region +2.6 [−0.5, +5.8], after verifying the region is causally live (3.32×, −10.5 pp masked). Vision-tower localisation: best single layer 1.6%, learned combination over 24 layers 1.0%, mean 0.5%, chance 2.3%.

K Probes and the Orgad port

Hidden states at five positions $\times$ 28 layers, OOF logistic probes, size-matched random-region control mandatory (without it a probe separates “annotated object” from “random rectangle” and once read AUROC 1.000; retracted). Target err: final position 0.714/0.657, evidence tokens 0.587/0.626, random region 0.503/0.516. Orgad et al. (2025)’s exact-answer-token claim does not port: evidence tokens are 0.127 and 0.031 worse than the final position. The token-level cliff (0.513 below vs 0.656 above on Qwen3-VL) did not replicate (0.623 vs 0.633 on Qwen2-VL) and is withdrawn. Evidence-token information peaks at L1 and decays to chance by L27 on Qwen3-VL. The pre-generation detector from the attention depth profile: AUROC 0.630 vs baselines 0.588/0.625 on Qwen3-VL, 0.579 vs 0.651/0.679 on Qwen2-VL; predictive entropy wins on both.

L Negatives in full

Window sizer. Pre-registered five-value grid, OOF per-item W from six free pass-1 features. Qwen3-VL +8.9 [+0.5, +17.3] over the bar on the five-value grid, inverting to

+6.8 [−1.6, +14.7] and −1.0 against the fixed constant on the three-value grid; Qwen2-VL on the pre-registered grid +3.7 [−3.1, +10.5] vs bar and −3.1 vs the constant, sending 43/191 items to the worst window. Dead. The `top1_frac` gate (+9.6 [+2.0, +17.7] on Qwen3-VL) was barred by pre-registration from substituting.

Multi-crop. $k=4$ windows at $B_0/4$ each in one pass: +7.9 pp over top-1 on V*Bench with the argmax proposer and +9.5 over the bar on single-region, transferring to 4K (+8.0) — but rank-only multi-crop with the head is −3.1 against a predicted +5.5 (prompting confound excluded), and the 4K win was measured with the crippled localiser (App. M).

Budget from a free signal. No pass-1 signal predicts the required budget: 9.4% exact-rung against a 35.6% majority baseline. peak predicts coverage at AUROC 0.788 but correctness at 0.50; random routing at the same cost already earns +3.3 pp from the concavity of the uniform curve, so peak’s own contribution is +1.9.

Top- k verification. The head’s top-2 recovers 14–17% of top-1 misses (+4.8/+5.9 coverage $\approx$ +3–4 pp accuracy) while a third pass moves the bar to uniform@900 ($\approx$ +4 pp); top-2 is adjacent to top-1 on 67% of items. Predicted null; not run. Misses are distractors (median 3–5 cells away, 5–8% near-misses), so spatial fixes cannot help.

Two refuted diagnostics. Both were proposed to explain a 4K failure that turned out to be an artefact. “Allocation wins where the budget axis has saturated”: V*Bench’s axis is near-linear (+8.0, +7.3, +8.7 pp per doubling). “The crop must be re-encoded at native resolution, so $W^* \approx 485/\text{side}$ ”: predicted 0.12 at 4K, measured 0.25, flat on single-object.

M Corrections made during this work

1. Every HR-Bench 4k phase before the final one localised on the bare question. Eight scripts, cloned from the first, passed the question with the options and the answer instruction stripped. On identical rows with both uniform arms agreeing on 100% of predictions, restoring the full prompt moves the crop arm by +19.3 pp and changes the argmax cell on 55% of items. Every 4K negative in an earlier version — the transfer negative, 4K gating, our arm in the 4K prior-art table, four “attempts to win at 4K”, the scale sweep, 4K multi-crop — is withdrawn; the prior-art losses stand (they run their own localisers). The scale boundary we spent five phases explaining was a prompt bug. Rule: never clone a localiser; import one.
2. The double-normalisation bug in our logit lens (App. A) invalidated three sections, including “DoLa/DeCo beaten by a free read-out”; corrected, no layer-decoding rule beats the model’s own logits.
3. An ROI-selection artefact produced a probe AUROC of 1.000 on the cohort the model answers wrong; the control that exposed it is now mandatory.
4. A capture-fraction mispairing would have inverted a conclusion (83% $\rightarrow$ 88% matched; the mismatched pairing reads 95%).
5. “Signed contrast is the mechanism” and “the final layer is anti-correlated” were Qwen3-VL-specific and are not claimed.
6. A void phase was built on the belief that the head read only L16–26; it read all 28.
7. A void run patched Qwen3-VL’s attention module while running Qwen2-VL; all arms identical, CI [+0.0, +0.0].
8. The token-level cliff (App. K) was called this paper’s best result before its second model; it failed there and is withdrawn. Four single-model results failed replication in one session; none is characterised as strong before its second model.
9. RePOPE: confident model disagreement is a $7.2\times$ enriched label-error detector, and every finding strengthens on the clean pool.

N Reproducibility

Code, per-phase scripts (phase00–phase142), the findings log with every number and its pre-registration, the replication ledger, and 91 invariant tests over the pipeline are released. Qwen3-VL-2B-Instruct, Qwen2-VL-7B-Instruct, LLaVA-NeXT-7B and LLaVA-OneVision-7B via HuggingFace transformers 5.3, bf16, eager attention where attention weights are read, one RTX 6000 Ada. Every figure in this paper is drawn from a stored forward pass on a real benchmark item.